



Converting fractions to terminating decimals

Task A: Recognising terminating and non-terminating

1) Without a calculator, calculate the decimal value of the following fractions:

a) $\frac{7}{20} = 0.35$

d) $\frac{47}{200} = 0.235$

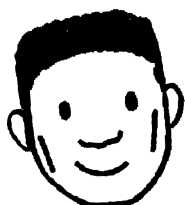
b) $\frac{5}{8} = 0.625$

e) $\frac{627}{50} = 12.54$

c) $\frac{3}{25} = 0.12$

f) $\frac{3}{16} = 0.1875$

Lucas says



$\frac{33}{132}$ is not equivalent to a **terminating decimal**.

$$\frac{33}{132} = \frac{33}{2 \times 2 \times 3 \times 11}$$

The prime factors of the denominator are 2, 3 and 11.

2) Explain why $\frac{33}{132}$ is equivalent to a **terminating decimal**.

Lucas needed to simplify the fraction $\frac{3 \times 11}{2 \times 2 \times 3 \times 11} = \frac{1}{2 \times 2} = \frac{1}{4}$

3) Jacob has discovered all these denominators will give a **terminating decimal**.

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \frac{1}{128}, \frac{1}{256}$$

Can you explain why?

All of the denominators only have a prime factor base of 2, so all are factors of powers of 10.

$$\frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \frac{1}{2^4}, \dots$$

4) Circle all the **terminating decimals**.

$$\begin{array}{ccccccc}
 \frac{2}{3} & \left(\frac{3}{2}\right) & \frac{4}{15} & \left(\frac{12}{15}\right) & \frac{7}{440} & \frac{11}{33} \\
 \frac{6}{7} & \left(\frac{11}{10}\right) & \frac{11}{12} & \left(\frac{7}{40}\right) & \frac{110}{330}
 \end{array}$$

Task B: Using a calculator

1) Using a calculator, match the fraction to its **terminating decimal**.

$$\begin{array}{cccc}
 0.1294 & 13.444 & 0.39484 & 0.856 \\
 \hline
 \frac{107}{125} & \frac{9871}{25\,000} & \frac{647}{5000} & \frac{3361}{250}
 \end{array}$$

(Note: Lines connect 0.1294 to 107/125, 13.444 to 9871/25000, 0.39484 to 647/5000, and 0.856 to 3361/250.)

2) Find the missing value(s) on the calculator displays.

Calculator display: 0.156 and $\frac{39}{250}$

Calculator display: 6.4595 and $\frac{12919}{2000}$

Calculator display: 0.2695 and $\frac{539}{2000}$

Calculator display: 15.245 and $\frac{3049}{200}$

